Beyond Popularity: A Bias-Controlled Re-evaluation of HMA-GCA for circRNA–miRNA Interaction Prediction

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Abstract

Circular RNAs (circRNAs) can bind microRNAs (miRNAs), and deep learning models are now widely used to predict these interactions. HMA-GCA [1], a recent model, reports AUC values of 0.963 to 0.973 on three datasets, but it uses random pair splits and random negative pairs, a setting in which a model can score well simply by learning which RNAs are popular. Here we test how much of this performance reflects real pair-level learning. We merged the three datasets into one benchmark of 3,802 circRNAs, 1,273 miRNAs and 20,771 unique positive pairs, built warm and cold-start splits with random and degree-matched negatives, and checked all splits for leakage. In the paper-style setting (warm split, random negatives), a rule that uses only node degree reached an AUC of 0.893, while HMA-GCA reached 0.958 and gradient boosting on sequence features reached 0.960. With degree-matched negatives, the degree rule fell to 0.494 and HMA-GCA to 0.817. Using node degree as a model input harmed prediction for unseen RNAs: when both the circRNA and the miRNA were new, AUC with random negatives was 0.546 with the degree input and 0.809 without it. Prediction for new miRNAs remained difficult (AUC 0.689 with degree-matched negatives), and gradient boosting performed as well as or better than HMA-GCA in this case. Removing the PCA and UMAP step had no consistent effect. These findings show that reported scores depend strongly on how the data are split and how negatives are chosen, and we recommend that future studies report degree-only baselines, cold-start splits, degree-matched negatives and node-held-out validation.

Keywords: circRNA–miRNA interaction, link prediction, benchmark audit, cold-start evaluation, node degree bias, negative sampling

1 Introduction

Circular RNAs (circRNAs) are closed-loop RNA molecules that can act as sponges for microRNAs (miRNAs) and so change how genes are regulated [2]. Large resources now collect such relations. For example, starBase v2.0 identified about 9,000 miRNA–circRNA relationships from CLIP-Seq data [3], and CircTarget collects experimentally supported circRNA–target RNA interactions from several cell types [4]. Testing a circRNA–miRNA pair in the laboratory is slow and costly, so computational models are used to rank candidate pairs before experiments.

Many deep learning models have been proposed for this task. DGCLCMI combines word2vec sequence embeddings with graph collaborative filtering and reports an average AUC of 0.960 on three datasets [5]. MFERL uses multi-scale representation learning with a dual-convolution attention module and contrastive learning [6]. iHofman joins high-order and low-order sequence features with

weighted attention and reports an AUC of 0.9092 on its baseline dataset [7]. HMA-GCA [1] is one of the most recent. It builds sequence and network descriptors for every RNA, appends PCA and UMAP projections of these descriptors ("hybrid manifold augmentation"), and joins the circRNA and the miRNA with a gated cross-attention module. It reports AUC values of 0.9713, 0.9630 and 0.9731 on three existing circRNA–miRNA datasets that its authors took from earlier work [6, 8, 9] (Section 3.1).

High scores on this kind of task need careful reading. A circRNA–miRNA network is sparse and uneven: a few RNAs have many known partners and most have very few. In such a network, a model can score well by learning that popular RNAs tend to interact, without learning anything about a specific pair. Studies in other fields show that this is a real risk. In protein–protein interaction prediction, random splits led to strongly overestimated performance, and models learned mainly from sequence similarity and node degree; when leakage was removed, performance became random [10]. Work on scale-free biomolecular networks found that network topology strongly affects how models are trained and judged, and proposed a degree-distribution-balanced sampling strategy for negatives [11]. In mRNA–protein interaction prediction, performance fell sharply when the test proteins were not seen in training [12]. A broad audit of biomolecular benchmarks found cross-split contamination and label conflicts that can bias benchmark metrics [13]. In biomedical knowledge graphs, a study of data leakage showed the impact of train–test redundancy, but found no evidence that node degree alone drove the predictions of the models it tested, and found that random and cold-start splits can overstate performance on an independent test set [14]. So the effect of popularity is not the same in every setting and must be measured for each task. In miRNA–disease association prediction, RGCNMDA reports its results separately for random, cold-disease and cold-miRNA evaluation, and its AUC falls from 0.9589 to 0.9109 and 0.8836 across these settings on HMDD v4.0 [15]. Finally, in a link-prediction study on biomedical graphs, simple baselines such as Common Neighbours were a justifiable choice in the smallest graph (about 15,000 edges) [16]. The merged circRNA–miRNA graph used here has a similar size (20,771 edges).

The authors of HMA-GCA are open about some of these limits. They write that random negative sampling may include undiscovered positives, and that their model shows popularity bias, favouring well-studied genes [1]. What is missing is a measurement of how large the effect is. This is the aim of the present study.

We ask three questions:

1. **How much of the paper-style score can be explained by popularity alone?**
2. **How does performance change when the test set uses degree-matched negatives, and when the circRNA, the miRNA or both are new (cold-start)?**
3. **Which parts of the HMA-GCA pipeline matter under these stricter conditions, in particular the node-degree input and the PCA+UMAP step?**

Our contributions are:

- A merged and deduplicated circRNA–miRNA benchmark built from the three existing circRNA–miRNA datasets used by HMA-GCA, with four split types, two kinds of test negatives and documented leakage checks.
- A comparison of a degree-only rule, gradient boosting and four variants of the HMA-GCA architecture on this benchmark, with three random seeds for the two main variants.
- Evidence that the node-degree input strongly hurts cold-start prediction, that the PCA+UMAP step has no consistent effect in our tests, and that prediction for new miRNAs is the main open problem.
- A comparison of validation protocols, showing that a random validation slice gives a misleading picture in cold-start settings.

We stress that this is not a reproduction of the published HMA-GCA numbers. We use only the K-mer and CTD descriptors, and our splits and negatives differ from those of the original study (Section 2.7).

2 Methodology

2.1 Overview

The study has four steps: (i) describe the source datasets and their popularity structure; (ii) merge the three datasets used by HMA-GCA into one benchmark with leak-free splits and two kinds of test

negatives; (iii) run baselines and HMA-GCA variants on exactly the same stored splits; (iv) compare models fold by fold. Every model reads the same stored files, so all models see identical data. Figure 1 summarises this workflow.

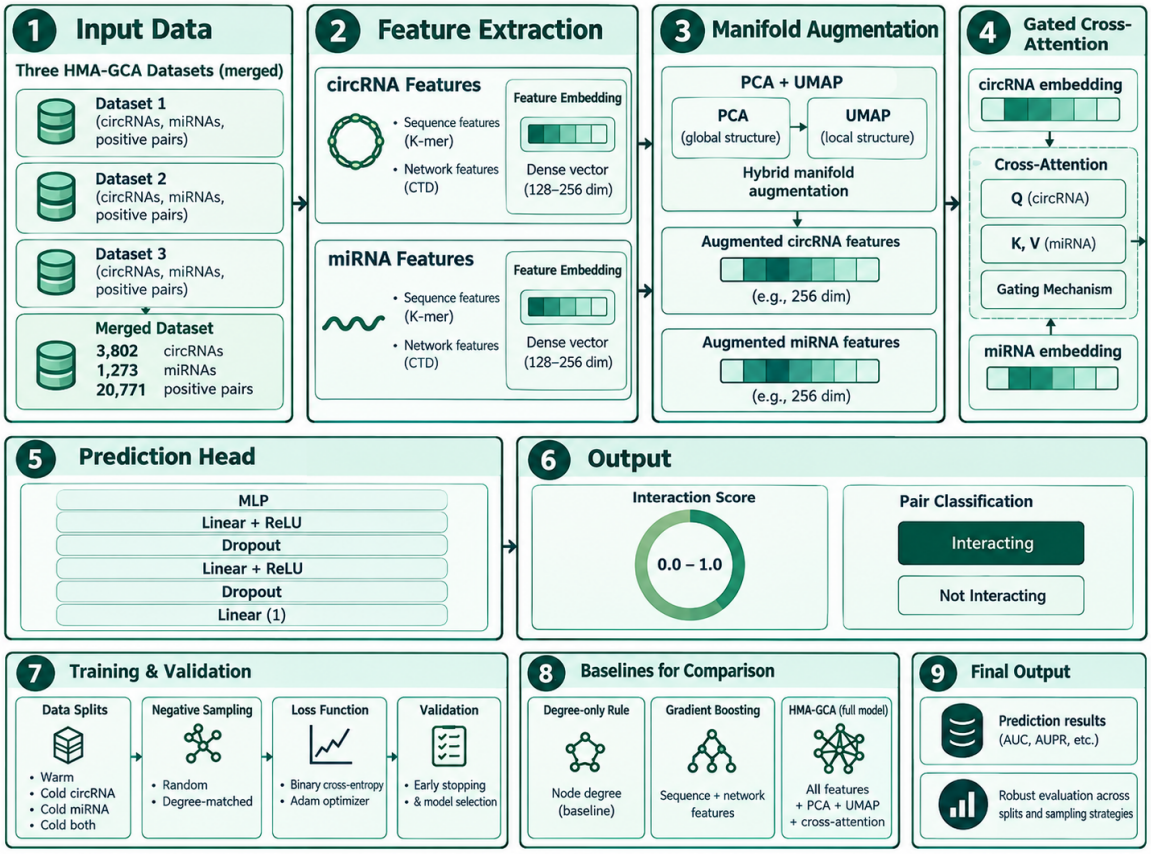


Fig. 1 Study workflow: from the three source datasets, through merging, splitting and negative sampling, to the baselines and HMA-GCA variants and the paired comparison.

2.2 Benchmark construction

Merging. The circRNAs and miRNAs of the three datasets used by HMA-GCA were combined by identifier. A pair is a unique (circRNA ID, miRNA ID) combination, and each pair keeps three flags that show which original datasets contain it, so each original dataset can be recovered. The K-mer and CTD vectors of an RNA were identical in every dataset in which it appears (no difference larger than 0.0001 in 4,228 circRNA overlaps and 1,662 miRNA overlaps), so the features could be merged without conflicts.

Splits. Each split has five folds, and the fold seed is 42.

Split	Definition	What it tests
Warm	20% of the positive pairs are held out in each fold (random pair-level split, as in the HMA-GCA paper)	Prediction for known RNAs
Cold circRNA	circRNAs are divided into five groups. All pairs of the test group's circRNAs are removed from training	New circRNAs
Cold miRNA	The same, by miRNA	New miRNAs
Cold both	Test pairs have both the circRNA and the miRNA in the test group. Training pairs have neither in the test group. Pairs with exactly one test endpoint are dropped	New circRNAs and new miRNAs together

Negatives. A negative pair is any pair that is not a known positive. The number of negatives always equals the number of positives (1:1).

- *Training negatives* are drawn uniformly at random from the nodes that are allowed in training.
- *Test negatives* come in two versions for the same test positives. *Random* negatives are drawn uniformly from the nodes allowed in the test fold, as in the HMA-GCA paper. *Degree-matched* negatives are drawn so that each endpoint is chosen with probability proportional to its degree in the full positive graph, restricted to the nodes allowed in the test fold. Negatives therefore have about the same popularity as the positives, and a popularity rule has no free advantage.
- Training and test negatives never overlap, and no negative is a known positive.

Leakage checks. For every split and fold (20 folds in total) we recorded four counts, and all were zero: positive pairs found in both training and test; negatives that are known positives; negatives shared between training and test; and (in cold splits) test nodes that also appear in training. We also re-derived the merged pairs and the splits from the raw matrices with separate code. The merged pair set equalled the union of the raw pairs (20,771), every positive row was a real positive, no negative row was a real positive, and no test positive appeared in training.

2.3 Features

Each RNA is described by two sequence-derived descriptors: K-mer frequencies (1-mer to 4-mer, 340 values) and CTD descriptors of composition, transition and distribution (30 values), 370 values in total. We did not use the Doc2Vec and Role2Vec descriptors of HMA-GCA. In an earlier check on the released files, these vectors for the same RNA were not the same in different datasets, so they cannot be merged, and Role2Vec is computed from the interaction network and may carry information about test edges. Our results therefore describe the HMA-GCA architecture and pipeline on reduced inputs.

2.4 Baselines

- **Popularity.** The score of a pair is $\log(1 + \text{degree of the circRNA}) + \log(1 + \text{degree of the miRNA})$. Degrees are counted from the training positives only, so a node not seen in training has degree 0.
- **Gradient boosting, sequence only.** A gradient boosting classifier (scikit-learn `HistGradientBoostingClassifier`, 150 iterations, otherwise default settings) on the circRNA and miRNA K-mer+CTD vectors joined together (740 values). It was not tuned.
- **Gradient boosting, sequence + degree.** The same, plus $\log(1 + \text{training degree})$ of both nodes.

2.5 HMA-GCA architecture and feature pipeline

Feature pipeline. For each RNA type separately, starting from the 370 K-mer+CTD values, we (1) computed the cosine-similarity matrix among all nodes of that type; (2) added the mean similarity and the mean of the five highest similarities as two columns; (3) added a 64-dimensional PCA of the similarity matrix; (4) added the node degree, counted from the training positives of the fold; (5) appended a 32-dimensional PCA and an 8-dimensional UMAP of the result (the "manifold augmentation"); and (6) standardised every column. The full pipeline gives 477 values per node (437 before the manifold step). The similarity, PCA and UMAP steps use the features of all nodes but no interaction labels, as in the original HMA-GCA setting. The hyper-parameters (PCA 32, UMAP 8,

hidden size 256, learning rate 5e-4, weight decay 5e-5, batch size 256, gradient clipping 5.0) follow the HMA-GCA paper [1].

Network. A separate encoder for each RNA type maps the input to 256 values: linear layer, batch normalisation, ReLU, dropout 0.3, and a second linear layer. A gated cross-attention module then computes a query from the circRNA embedding, and a key and a value from the miRNA embedding. The score is the sum over all 256 channels of the product of query and key, divided by $\sqrt{256}$. This is one number per pair. It passes through a sigmoid and scales the miRNA value vector. Finally, the circRNA embedding and the gated miRNA vector are joined (512 values) and passed to a predictor (linear 256, ReLU, dropout 0.4, linear 1). Our implementation differs from the equations of the paper in two small ways: the encoder has a second linear layer, and the joined vector goes to the predictor without a LayerNorm layer (Eq. 15 of the paper includes one). We used the model code of the authors' public repository.

Variants.

Variant	Degree input	PCA+UMAP appended	Purpose
Full	yes	yes	The pipeline as published
Without degree	no	yes	Does the model depend on the popularity shortcut?
Without PCA+UMAP	yes	no	Effect of the manifold step (mixed with the degree input)
Without degree and without PCA+UMAP	no	no	Clean test of the manifold step

2.6 Training and validation protocol

All models use binary cross-entropy, the Adam optimiser and cosine annealing of the learning rate over 50 epochs. Validation AUC is computed every two epochs, training stops after eight checks without improvement (16 epochs), and the best weights are restored. The test folds are never used for model selection. We ran this in two rounds.

- **Round 1 (v1).** A random 10% of the training pairs formed the validation set in all splits, with one seed (42), for three variants (full, without degree, without PCA+UMAP).
- **Round 2 (v2, the main results).** The validation set matches the test condition. In the warm split it is a random 10% of the training pairs. In the cold circRNA split, 10% of the training circRNAs are held out together with all their pairs; in the cold miRNA split, 10% of the miRNAs; and in the cold both split, 15% of the circRNAs and 15% of the miRNAs. Node degrees are counted after the validation pairs are removed. All four variants were run with seed 42 (five folds each). The full and the without-degree variants were also run with seeds 43 and 44. In total, v2 contains 160 training runs, each evaluated on both kinds of test negatives.

The gradient-boosting baselines were trained on all training rows of each fold, without a validation set, so in the cold splits they used more training data than the v2 neural models.

2.7 Metrics, comparisons and differences from the original study

We report the area under the ROC curve (AUC), computed on each test fold and then averaged; the area under the precision–recall curve was also recorded and is available in the result files. Models are compared on the same folds. For two models we compute the difference in AUC on each fold (and each seed, when available) and report the mean, the standard deviation of the differences and the number of pairs that favour the first model. We did not apply formal significance tests, because runs with different seeds share the same folds and are not independent.

Our setting differs from the original HMA-GCA study in three ways: (i) only K-mer and CTD inputs are used; (ii) early stopping uses a held-out validation set, not the evaluation fold; and (iii) splits and negatives come from the merged benchmark, including cold-start splits and degree-matched negatives. The HMA-GCA paper uses five-fold cross-validation with an interaction-pair-level split, random negatives and an early-stopping patience of eight epochs [1].

2.8 Dataset-level popularity analysis

Separately from the models, we measured how much popularity alone explains for each dataset. A pair is scored by (RNA degree) × (miRNA degree). Positives are split into five folds, the test positives are hidden from the graph, degrees are counted on the remaining graph, and the AUC is computed against random or degree-matched negatives. In this protocol, AUC values below 0.5 can occur under degree-matched negatives, because hiding a positive edge lowers the degree of its own two endpoints while the negatives are unaffected. We read such values as "popularity gives no signal", not as a real signal. This analysis uses its own protocol, so it shows how strong the shortcut is in each dataset but is not a re-measurement of the original pipelines.

2.9 Implementation

All code was written in Python with scikit-learn, PyTorch and umap-learn, and was run in Kaggle notebooks with a GPU. The seed was 42 for data construction. The notebooks and result files are listed in the Data and Code Availability section.

3 Datasets

3.1 Source datasets

HMA-GCA did not create these datasets. Its authors selected three existing circRNA–miRNA datasets from earlier work: one obtained from circBank, which they cite to [6]; one from BGF-CMA [8]; and one from CMI data used in [9]. They labelled these Dataset 1, 2 and 3 [1] and release the files in their code repository. We keep their numbering, so "Dataset 1–3" in this paper means the datasets as numbered and released by HMA-GCA, and the data credit belongs to the original sources. We used the released files and did not rebuild the datasets from the original sources. Table 1 gives the counts, measured from the files, together with the merged benchmark.

Table 1 Size, sparsity and degree structure of the datasets. Datasets 1–3 are the three datasets used by HMA-GCA [1], numbered as in that paper: Dataset 1 is from circBank (cited by HMA-GCA to [6]), Dataset 2 is from BGF-CMA [8], and Dataset 3 is the CMI data of [9]. We used the files as released by the HMA-GCA authors (Section 3.1). Density is the share of all possible pairs that are known positives. The Gini coefficient measures how unevenly the edges are spread over nodes (0 = even, 1 = all edges on one node).

Dataset	RNA type	RNAs	miRNAs	Positive pairs	Density (%)	Mean RNA degree	Mean miRNA degree	Gini (RNA)	Gini (miRNA)
Dataset 1	circRNA	2,115	821	9,589	0.552	4.53	11.68	0.560	0.692
Dataset 2	circRNA	2,346	962	9,905	0.439	4.22	10.30	0.564	0.704
Dataset 3	circRNA	3,569	1,152	20,208	0.491	5.66	17.54	0.564	0.725
Merged benchmark (this work)	circRNA	3,802	1,273	20,771	0.429	5.46	16.32	0.572	0.732

All counts of Datasets 1–3 match Table 1 of the HMA-GCA paper, except one: the files of Dataset 2 contain 962 miRNAs, whereas the paper reports 926. We do not know the cause (a typing error in the paper is possible, but we have not confirmed it). Our results use the file contents.

3.2 The three circRNA datasets overlap

Datasets 1, 2 and 3 are nested, not independent (Table 2). Every pair of Dataset 1 is also in Dataset 2. Of the pairs of Dataset 1, 247 are not in Dataset 3, and of the pairs of Dataset 2, 563 are not in Dataset 3 (247 + 316). Of the 20,208 pairs of Dataset 3, 10,866 are in neither of the others. As a result, training on one dataset and testing on another would leak pairs. This is one reason why we merge the datasets and build new splits.

Table 2 Number of unique pairs by the combination of source datasets that contain them. Datasets 1–3 are as in Table 1.

Pair found in	Number of unique pairs
Dataset 3 only	10,866
Datasets 1, 2 and 3	9,342
Dataset 2 only	316
Datasets 1 and 2, not 3	247

3.3 The merged benchmark

The merged benchmark contains 3,802 circRNAs, 1,273 miRNAs and 20,771 unique positive pairs. Its density is 0.429%. Each RNA has 370 K-mer+CTD values (Section 2.3). The K-mer part of the miRNA descriptor is mostly empty: 86.7% of the miRNA K-mer values are zero, compared with 3.8% for circRNAs. A miRNA of about 22 nucleotides cannot fill 340 K-mer dimensions, so the miRNA representation is weak. This matters for the results on new miRNAs (Section 4.5).

3.4 Sparsity and popularity structure

All datasets are sparse (density 0.43%–0.55%) and uneven (Table 1, Table 3, and Figure 2). In the merged benchmark, the median circRNA has 3 known partners and the median miRNA has 4, while the most connected circRNA has 74 and the most connected miRNA has 420. About a third of the circRNAs (31.1%) and a quarter of the miRNAs (25.1%) have a single known partner. In Datasets 1–3, the top 10% of the circRNAs hold 43.0%–44.3% of the edges and the top 10% of the miRNAs hold 59.0%–61.2%.

Table 3 Degree statistics of the circRNA datasets. Datasets 1–3 are as in Table 1.

Dataset	circRNA median degree	circRNA max degree	circRNAs with degree 1 (%)	miRNA median degree	miRNA max degree	miRNAs with degree 1 (%)	Edges held by top 10% circRNAs (%)	Edges held by top 10% miRNAs (%)
Dataset 1	2	73	39.1	4	181	23.4	43.0	59.0
Dataset 2	2	73	43.4	3	181	28.9	44.3	61.2
Dataset 3	3	73	28.2	5	420	22.7	43.3	60.9
Merged	3	74	31.1	4	420	25.1	44.4	61.9

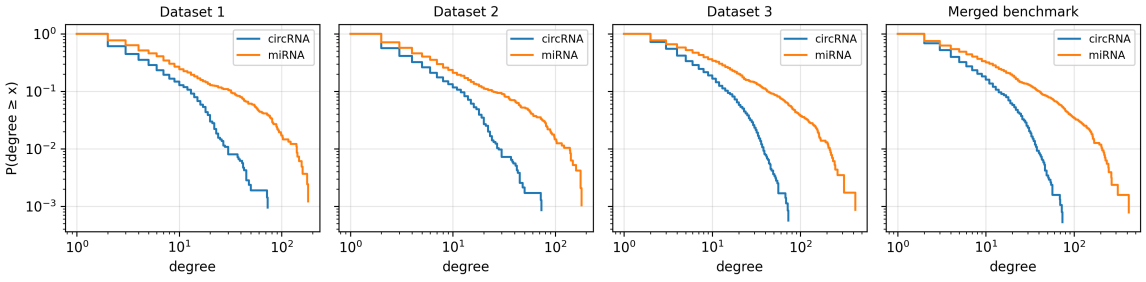


Fig. 2 Degree distributions (complementary cumulative distribution, log–log) of Datasets 1–3 and the merged benchmark. Long straight tails show that a few RNAs hold many edges.

3.5 Split sizes and test-set composition

The cold both folds are small (738–960 test positives), so results in this split are noisier.

Table 4 Number of positive pairs per fold. Each test set has the same number of negatives.

Split	Training positives per fold	Test positives per fold	Mean test positives
Warm	16,616–16,617	4,154–4,155	4,154
Cold circRNA	16,232–16,908	3,863–4,539	4,154
Cold miRNA	16,352–17,208	3,563–4,419	4,154
Cold both	12,826–13,700	738–960	831

Table 5 Mean degree of the test pairs in the full positive graph (all folds pooled). Random negatives are much less popular than positives; degree-matched negatives are close to the positives.

Split	Positives (circRNA / miRNA degree)	Random negatives	Degree-matched negatives
Warm	16.1 / 96.0	5.5 / 15.7	15.5 / 90.9
Cold circRNA	16.1 / 96.0	5.4 / 16.2	15.1 / 88.7
Cold miRNA	16.1 / 96.0	5.6 / 16.3	15.1 / 87.2
Cold both	16.2 / 94.8	5.5 / 15.8	15.1 / 87.4

Degree matching is by endpoint (each node is drawn by its own degree), not by pair, and the matched negatives are still slightly less popular than the positives (for example, 90.9 against 96.0 for miRNAs in the warm split).

4 Results

4.1 Popularity alone explains much of the score under random negatives

Table 6 shows the dataset-level popularity analysis (Section 2.8). On the three circRNA datasets, the degree product reaches AUC 0.800–0.847 with random negatives, and 0.457–0.483 with degree-matched negatives, so the shortcut disappears when the negatives are as popular as the positives. The miRNA degree alone is more informative (0.823–0.852) than the circRNA degree alone (0.712–0.730).

Table 6 AUC of a learning-free popularity rule on each dataset (five folds).

Dataset	Popularity AUC, random negatives	Popularity AUC, degree-matched negatives	RNA degree only (random negatives)	miRNA degree only (random negatives)
Dataset 1	0.808	0.457	0.714	0.826
Dataset 2	0.800	0.466	0.712	0.823
Dataset 3	0.847	0.483	0.730	0.852

4.2 The benchmark passes all integrity checks

All leakage counts were zero in all 20 folds, and the independent re-derivation from the raw matrices agreed with the merged benchmark (Section 2.2). In the cold both split, the popularity rule scores exactly 0.500 because both nodes of every test pair have degree 0 and all scores tie.

4.3 Main results

Tables 7 and 8 show the mean AUC for all models, splits and test negatives (HMA-GCA v2). The number after ± is the standard deviation across folds (for the full and without-degree variants: across 15 seed–fold runs; for the other variants and baselines: across five folds).

Table 7 Mean AUC ± SD with **random** test negatives (the setting of the HMA-GCA paper).

Model	Warm	Cold circRNA	Cold miRNA	Cold both
Popularity (degree only)	0.893 ± 0.002	0.853 ± 0.004	0.730 ± 0.006	0.500 ± 0.000
Gradient boosting, sequence only	0.960 ± 0.001	0.955 ± 0.002	0.881 ± 0.008	0.873 ± 0.017
Gradient boosting, sequence + degree	0.900 ± 0.003	0.911 ± 0.004	0.706 ± 0.021	0.676 ± 0.040
HMA-GCA full	0.958 ± 0.002	0.937 ± 0.004	0.721 ± 0.027	0.546 ± 0.043
HMA-GCA without degree	0.967 ± 0.002	0.961 ± 0.003	0.820 ± 0.022	0.809 ± 0.013
HMA-GCA without PCA+UMAP	0.965 ± 0.002	0.950 ± 0.004	0.804 ± 0.026	0.721 ± 0.025
HMA-GCA without degree and without PCA+UMAP	0.966 ± 0.002	0.960 ± 0.003	0.830 ± 0.010	0.795 ± 0.022

Table 8 Mean AUC ± SD with **degree-matched** test negatives.

Model	Warm	Cold circRNA	Cold miRNA	Cold both
Popularity (degree only)	0.494 ± 0.006	0.517 ± 0.007	0.488 ± 0.004	0.500 ± 0.000
Gradient boosting, sequence only	0.783 ± 0.003	0.792 ± 0.007	0.692 ± 0.019	0.691 ± 0.024
Gradient boosting, sequence + degree	0.673 ± 0.006	0.700 ± 0.016	0.596 ± 0.031	0.688 ± 0.040
HMA-GCA full	0.817 ± 0.011	0.842 ± 0.012	0.590 ± 0.034	0.556 ± 0.037
HMA-GCA without degree	0.863 ± 0.009	0.857 ± 0.009	0.689 ± 0.036	0.680 ± 0.022
HMA-GCA without PCA+UMAP	0.817 ± 0.011	0.832 ± 0.013	0.658 ± 0.035	0.640 ± 0.046
HMA-GCA without degree and without PCA+UMAP	0.865 ± 0.006	0.859 ± 0.010	0.699 ± 0.022	0.671 ± 0.040

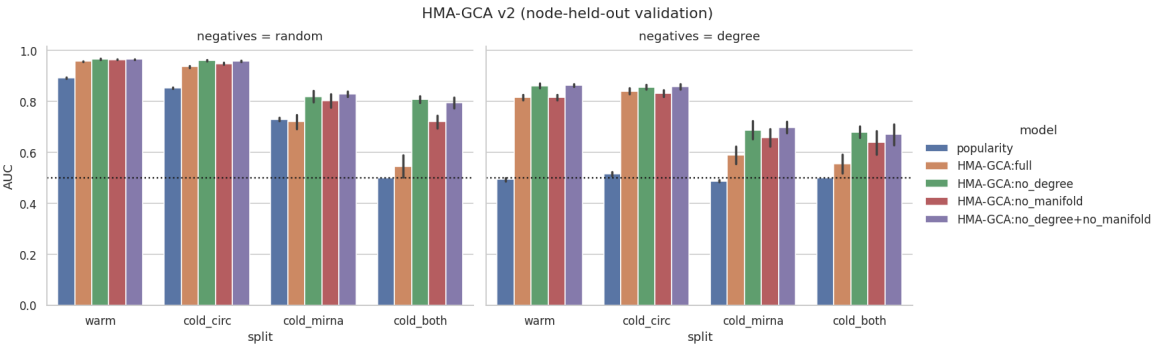


Fig. 3 Mean AUC (bars) and SD (lines) by split for the popularity rule and the four HMA-GCA variants, with random negatives (left) and degree-matched negatives (right).

Five points stand out.

1. **The paper-style score is reproduced with reduced inputs.** In the warm split with random negatives, HMA-GCA full reaches 0.958 and the variant without degree reaches 0.967, using only K-mer and CTD features. The published values on the three datasets are 0.9713, 0.9630 and 0.9731 [1]. Our numbers come from a different, merged benchmark and are indicative only.
2. **A simple baseline reaches the same level.** Gradient boosting on sequence features alone reaches 0.960 in this setting, and the degree-only rule reaches 0.893.
3. **Performance falls under stricter evaluation.** Table 9 shows the drop from the paper-style setting to the hardest setting. For the variant without degree, AUC falls from 0.967 to 0.863 when only the negatives are made degree-matched, and to 0.680 when both nodes are also new.
4. **The full model suffers most in cold-start.** With random negatives it falls to 0.721 for new miRNAs and 0.546 for new circRNA–miRNA pairs, close to chance.
5. **New circRNAs are much easier than new miRNAs.** In the cold circRNA split the variant without degree keeps 0.961 (random) and 0.857 (degree-matched). In the cold miRNA split it falls to 0.820 and 0.689.

Table 9 Mean AUC in three settings, and the total drop from the first to the third.

Model	Warm, random (paper-style)	Warm, degree-matched	Cold both, degree-matched (hardest)	Total drop
Popularity (degree only)	0.893	0.494	0.500	0.393
Gradient boosting, sequence only	0.960	0.783	0.691	0.269
Gradient boosting, sequence + degree	0.900	0.673	0.688	0.212
HMA-GCA full	0.958	0.817	0.556	0.402
HMA-GCA without degree	0.967	0.863	0.680	0.287

4.4 The degree input harms cold-start prediction

Removing the degree input improves HMA-GCA in every setting (Table 10). The gain is small in the warm split with random negatives (+0.008) and large for new miRNAs (+0.099 for both kinds of negatives) and for new circRNA–miRNA pairs (+0.262 with random negatives, +0.124 with degree-matched negatives). In 7 of the 8 settings all 15 seed–fold pairs favour the variant without degree; the exception is the cold circRNA split with degree-matched negatives (13 of 15).

Table 10 AUC of HMA-GCA **without degree** minus HMA-GCA **full** (15 seed–fold pairs per row).

Split	Test negatives	Mean AUC difference	SD of difference	Pairs favouring first model
Warm	Random	+0.008	0.003	15 of 15
Warm	Degree-matched	+0.046	0.011	15 of 15
Cold circRNA	Random	+0.023	0.005	15 of 15
Cold circRNA	Degree-matched	+0.016	0.011	13 of 15
Cold miRNA	Random	+0.099	0.035	15 of 15
Cold miRNA	Degree-matched	+0.099	0.033	15 of 15
Cold both	Random	+0.262	0.048	15 of 15
Cold both	Degree-matched	+0.124	0.041	15 of 15

The training data show a possible reason. In the training positives, every node has a training degree of at least 1, because the edge itself is counted. Training negatives are random, so many involve nodes without any training edge. In fold 0 of the warm split (degrees counted from all training positives), 6.9% of the training negatives have a circRNA with degree 0, and 4.8% have a miRNA with degree 0, against 0% of the training positives. The mean training degree of the circRNA is 13.1 for positives and 4.4 for negatives. In the cold splits, every test node on the cold side has degree 0 (100% in fold 0). A model that has learned "degree 0 means negative" will therefore score cold test pairs too low. This pattern agrees with the results, since the loss is largest in the cold both split. We did not test the explanation directly (for example, by replacing the test degree with a neutral value). The degree is also a problem in the warm split: with degree-matched negatives, the full model is 0.046 below the variant without degree (15 of 15 pairs).

4.5 Comparison with gradient boosting

Table 11 compares HMA-GCA without degree with gradient boosting on sequence features, fold by fold (HMA-GCA averaged over the three seeds).

When the miRNA is known (warm and cold circRNA splits), HMA-GCA without degree is better with degree-matched negatives (+0.079 and +0.065, five folds of five). With random negatives the advantage is small (+0.007 and +0.006) but consistent. When the miRNA is new, the picture changes. With random negatives, gradient boosting is clearly better (−0.061 and −0.064, no fold in favour of HMA-GCA). With degree-matched negatives, the two are equal within the spread of the folds (−0.003 and −0.011). So under the strictest conditions, the neural model does not show a clear gain over a simple tree model for new miRNAs.

Two cautions apply. First, the gain for known miRNAs cannot be assigned to the attention architecture alone, because HMA-GCA also receives similarity-based inputs (mean similarity, top-five

Table 11 AUC of HMA-GCA **without degree** minus gradient boosting (sequence only), five folds.

Split	Test negatives	Mean AUC difference	SD of difference	Folds favouring HMA-GCA
Warm	Random	+0.007	0.002	5 of 5
Warm	Degree-matched	+0.079	0.009	5 of 5
Cold circRNA	Random	+0.006	0.002	5 of 5
Cold circRNA	Degree-matched	+0.065	0.008	5 of 5
Cold miRNA	Random	−0.061	0.012	0 of 5
Cold miRNA	Degree-matched	−0.003	0.020	2 of 5
Cold both	Random	−0.064	0.006	0 of 5
Cold both	Degree-matched	−0.011	0.009	0 of 5

similarity and a PCA of the similarity matrix) that gradient boosting did not receive. Second, the gradient-boosting model was untuned and used all training rows, while the HMA-GCA runs used a smaller training set after the validation split.

4.6 The PCA+UMAP step shows no consistent effect

The clean test is the comparison of the variant without degree and without PCA+UMAP against the variant without degree (Table 12; five folds, seed 42). The mean differences range from −0.011 to +0.014. The signs are mixed, the number of folds in favour is between 1 and 4 of 5, and the standard deviation of the differences reaches 0.061 (cold miRNA, degree-matched). We therefore find no consistent benefit of the PCA+UMAP step in our setting.

Table 12 AUC of HMA-GCA **without degree and without PCA+UMAP** minus HMA-GCA **without degree** (five folds, seed 42).

Split	Test negatives	Mean AUC difference	SD of difference	Pairs favouring first model
Warm	Random	−0.001	0.002	2 of 5
Warm	Degree-matched	+0.002	0.007	3 of 5
Cold circRNA	Random	−0.001	0.001	1 of 5
Cold circRNA	Degree-matched	+0.005	0.005	4 of 5
Cold miRNA	Random	+0.008	0.020	4 of 5
Cold miRNA	Degree-matched	+0.014	0.061	4 of 5
Cold both	Random	−0.011	0.012	1 of 5
Cold both	Degree-matched	−0.005	0.039	3 of 5

The comparison of the variant without PCA+UMAP against the full model (Table 13) looks different: removing the manifold step seems to help in the cold splits with random negatives (for example +0.167 for both nodes new). This comparison is mixed with the degree input, since both variants keep it. One possible reading is that the manifold step makes the degree problem worse, because the step is fitted on features that include the degree. We did not test this.

4.7 Validation must match the test condition

In round 1, the random validation slice gave a validation AUC of 0.965–0.970 for every split, while the test AUC for the full model was 0.542 for both nodes new (Table 14). A random slice contains known nodes and therefore looks like the warm split, so early stopping did not reflect cold-start performance. With the node-held-out validation of round 2, validation AUC and test AUC agree closely (for example 0.564 and 0.546 for the full model with both nodes new). The agreement is weaker for the variant without degree and new miRNAs (0.844 against 0.820).

4.8 Stability across seeds and training length

Table 15 lists the mean AUC for each seed. The seed means of the variant without degree differ by at most 0.017 (cold miRNA, degree-matched: 0.682–0.699). For the full model, the seed means differ

Table 13 AUC of HMA-GCA **without PCA+UMAP** minus HMA-GCA **full** (five folds, seed 42). The degree input is present in both.

Split	Test negatives	Mean AUC difference	SD of difference	Pairs favouring first model
Warm	Random	+0.007	0.002	5 of 5
Warm	Degree-matched	+0.002	0.003	3 of 5
Cold circRNA	Random	+0.012	0.002	5 of 5
Cold circRNA	Degree-matched	−0.015	0.015	1 of 5
Cold miRNA	Random	+0.081	0.027	5 of 5
Cold miRNA	Degree-matched	+0.083	0.019	5 of 5
Cold both	Random	+0.167	0.055	5 of 5
Cold both	Degree-matched	+0.071	0.070	4 of 5

Table 14 Mean validation AUC and test AUC (random negatives) in the two rounds.

Protocol	Variant	Split	Validation AUC	Test AUC (random negatives)
v1 (random validation slice)	full	Warm	0.968	0.963
v1 (random validation slice)	full	Cold circRNA	0.967	0.939
v1 (random validation slice)	full	Cold miRNA	0.970	0.707
v1 (random validation slice)	full	Cold both	0.966	0.542
v1 (random validation slice)	without degree	Warm	0.966	0.966
v1 (random validation slice)	without degree	Cold circRNA	0.965	0.960
v1 (random validation slice)	without degree	Cold miRNA	0.969	0.830
v1 (random validation slice)	without degree	Cold both	0.965	0.820
v2 (node-held-out validation)	full	Warm	0.960	0.958
v2 (node-held-out validation)	full	Cold circRNA	0.938	0.937
v2 (node-held-out validation)	full	Cold miRNA	0.724	0.721
v2 (node-held-out validation)	full	Cold both	0.564	0.546
v2 (node-held-out validation)	without degree	Warm	0.968	0.967
v2 (node-held-out validation)	without degree	Cold circRNA	0.961	0.961
v2 (node-held-out validation)	without degree	Cold miRNA	0.844	0.820
v2 (node-held-out validation)	without degree	Cold both	0.815	0.809

by at most 0.030 (cold miRNA, degree-matched: 0.575–0.605). In the setting with the smallest gap between the full and the without-degree variants (cold circRNA, degree-matched), the seed means of the full model (0.836–0.847) are close to those of the other variant (0.855–0.859), and 13 of 15 pairs favour the variant without degree (Table 10). All other gaps in Table 10 are larger than the seed spread.

Table 15 Mean AUC per seed (five folds each).

Variant	Split	Random negatives (seed 42 / 43 / 44)	Degree-matched negatives (seed 42 / 43 / 44)
full	Warm	0.958 / 0.959 / 0.958	0.815 / 0.820 / 0.815
full	Cold circRNA	0.938 / 0.937 / 0.937	0.847 / 0.836 / 0.842
full	Cold miRNA	0.723 / 0.723 / 0.718	0.575 / 0.590 / 0.605
full	Cold both	0.554 / 0.543 / 0.543	0.569 / 0.547 / 0.554
without degree	Warm	0.967 / 0.966 / 0.967	0.863 / 0.860 / 0.866
without degree	Cold circRNA	0.961 / 0.961 / 0.961	0.855 / 0.857 / 0.859
without degree	Cold miRNA	0.822 / 0.824 / 0.814	0.685 / 0.699 / 0.682
without degree	Cold both	0.806 / 0.808 / 0.811	0.675 / 0.679 / 0.686

Training length also changed between the rounds. In round 1 the median run used the full 50 epochs, and 56.7% of runs reached the cap. In round 2 the median was 39 epochs (range 18–50), and 15.6% of runs reached the cap.

5 Discussion

On our merged benchmark, a model with fewer inputs and a simple gradient boosting classifier both reach the level of the published HMA-GCA scores, and these scores fall sharply once the evaluation removes shortcuts. In the paper-style setting, a degree-only rule already scores 0.893. With degree-matched negatives, this rule falls to chance (0.494), while HMA-GCA keeps 0.817–0.863, so the model does learn something beyond popularity. However, when the miRNA or both RNAs are new, HMA-GCA loses a further 0.17–0.26 AUC under degree-matched negatives. This is consistent with findings for unseen proteins in other interaction tasks [10, 12]. Our results do not show that HMA-GCA is wrong. They show that a single number, obtained from random pairs and random negatives, describes only one of the situations in which such a model is used. The HMA-GCA authors also name popularity bias and random negative sampling as limits of their work [1], and our results measure the size of these effects.

Node degree is easy to compute and strongly predictive when negatives are random, which makes it tempting to use as an input. When degree is computed from the training graph, however, it exists for known nodes and disappears for new ones. The model learns to rely on it and then fails where prediction is most useful, namely for RNAs without known partners. The training data suggest a likely mechanism: every training positive has a degree of at least 1, whereas many random negatives involve nodes with no training edge, so a model can learn that a degree of 0 means a negative pair (Section 4.4). We did not test this explanation by intervention. The same concern applies to any feature built from training labels, and it relates to the leakage-controlled feature construction of RGCNMDA, which rebuilds interaction profiles inside each fold from training positives only [15]. The effect of degree also depends on the task, since a permutation study in knowledge-graph link prediction found no evidence that degree alone drove predictions [14]. For circRNA–miRNA data, where the popularity rule alone reaches 0.893, the effect must be measured rather than assumed.

The paper presents PCA and UMAP as a key part of the model, but in our tests removing them changed AUC by between -0.011 and $+0.014$ with no consistent direction. The ablations described in the original paper cover the CTD and K-mer features, the language-model features and the attention mechanism, so the removal of PCA and UMAP was not isolated there [1]. Our result rests on one seed, five folds and K-mer and CTD inputs only, so it shows no benefit in our setting rather than proof that none exists. When the degree input was kept, removing the manifold step appeared to help in the cold splits. One possible reading is that the step, which is fitted on features that include degree, makes the degree problem worse, but we did not test this.

New miRNAs remain the main open problem. For these, gradient boosting performed as well as or better than HMA-GCA. A likely reason is that the miRNA description is weak, because 86.7% of the miRNA K-mer values are zero; we did not test this interpretation. Features of the seed region or embeddings from sequence language models would be natural improvements, but they need raw miRNA sequences, which the released HMA-GCA feature files do not contain. Validation also matters in this setting. A random validation slice contains known nodes and looks like the warm split, so in our first round it gave a validation AUC of about 0.97 where the test AUC was 0.54. Holding out whole nodes for validation removed this gap, and we suggest that cold-start studies do the same.

Based on these results, we recommend that future circRNA–miRNA studies report a degree-only baseline and results for warm, cold circRNA, cold miRNA and cold both splits. They should evaluate with degree-matched negatives in addition to random negatives, use node-held-out validation for cold-start settings, and avoid feeding training-set degree to the model unless its behaviour on unseen nodes has been tested. Finally, they should report several seeds with paired differences and compare against simple, strong baselines [10, 16].

5.1 Limitations

Several limits should be kept in mind when reading our results. We used only K-mer and CTD features and left out Doc2Vec and Role2Vec (Section 2.3), so this study is not a reproduction of the published numbers and does not use the authors' original splits. We re-ran only the HMA-GCA architecture, and we did not ablate its gated cross-attention module; other circRNA–miRNA models, such as DGCLCMI [5], MFERL [6] and iHofman [7], were not tested, so we cannot say whether they share these weaknesses. All models were trained on random negatives and tested on both kinds, so the drop under degree-matched test negatives mixes a harder test with a shift between training and test conditions. In addition, the degree matching is done for each endpoint and not for each positive pair, and the matched negatives are slightly less popular than the positives (Table 5). The

comparison with gradient boosting is not fully balanced, because the baseline was not tuned and used all training rows, whereas the neural models used fewer rows after the validation split. Only the full and without-degree variants were run with three seeds, so the comparisons in Tables 12 and 13 are weaker, and the cold both folds are small (738–960 test positives and 474–643 validation rows), which makes results there noisy. The similarity, PCA and UMAP steps use the features of all nodes, including test nodes, as in the original code, although they use no interaction labels. Our explanation of the degree failure agrees with the data but was not tested by an intervention (Section 4.4). The labels themselves also carry uncertainty. Unlabelled pairs are not true negatives, and some may be real but undiscovered interactions, as the HMA-GCA authors also note [1]. Public circRNA–miRNA resources can mix experimental evidence with computational predictions [2], and we did not check how the labels of Datasets 1–3 were produced in their original sources (Section 3.1); if some labels came from prediction tools, a model could partly learn the rules of those tools. The miRNA count of Dataset 2 in the released files (962) also differs from the published value (926), and we do not know why. Finally, we studied only circRNA–miRNA data. As a side check, we applied the popularity rule of Section 2.8 to the lncRNA–miRNA datasets released with DeepLMI [17] and LMI-MHGAT [18]. With random negatives it reached an AUC of 0.80 on both LMI-MHGAT datasets and 0.56 on the DeepLMI dataset, and with degree-matched negatives it gave 0.50–0.53 and 0.36, respectively (values below 0.5 are read as "no signal"). The shortcut therefore seems strong in some lncRNA–miRNA datasets and weak in others, but these numbers come from our own protocol, are not shown in the tables, and are not a re-measurement of either model, so we cannot say whether their reported scores contain a popularity effect.

5.2 Future work

The most direct next step is to apply the same audit to other models. DGCLCMI, MFERL and iHofman could be tested on our benchmark in the same way, and the lncRNA–miRNA predictors DeepLMI [17] and LMI-MHGAT [18] could be examined with a degree-only baseline and degree-matched test negatives, and, for LMI-MHGAT, with cold-start splits. DeepLMI already provides blind-lncRNA, blind-miRNA and blind-both splits, which allows a direct comparison with our cold-start results. Two changes to the evaluation would also make it stronger: training on degree-matched negatives, and matching degrees for each positive pair instead of each endpoint. Finally, better miRNA representations built from raw sequences should be tested, since new miRNAs are the hardest case. The lncRNA–miRNA data released with DeepLMI include raw miRNA sequences, and it remains to be tested whether miRNAs shared between circRNA and lncRNA data allow transfer between the two tasks.

6 Conclusion

We merged the three datasets used by HMA-GCA into a benchmark of 20,771 unique circRNA–miRNA pairs with cold-start splits, degree-matched negatives and documented leakage checks, and used it to re-evaluate the HMA-GCA architecture on K-mer and CTD features. In the paper-style setting, a degree-only rule reaches AUC 0.893, and both HMA-GCA (0.958) and gradient boosting (0.960) reach levels close to the published ones. With degree-matched negatives, HMA-GCA scores fall to 0.82–0.86 when the miRNA is known and to 0.56–0.70 when the miRNA is new, and the node-degree input makes cold-start prediction much worse. The PCA+UMAP step shows no consistent effect, and for new miRNAs a simple gradient boosting model is as good as or better than the neural model. These results show that reported performance depends on the evaluation protocol, and that predicting interactions for new miRNAs is the main open problem. Future work should test other models on the same benchmark, use richer miRNA representations, train with degree-matched negatives and check the origin of the interaction labels.

Data and Code Availability

The analysis used only the three public circRNA–miRNA datasets as released with HMA-GCA [1]; their original sources are named in Section 3.1. The notebooks (dataset analysis, benchmark construction and baseline audit, and the two HMA-GCA rounds) and their result files are available from the authors on request [add repository link before submission].

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